

SAT Math Mastery · Week 8B**Circles: Equations, Arcs, Tangents & Radians — Answer Key & Worked Solutions****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** circle equation · completing the square · center/radius from expanded form · circle position tests · arc length & sector area · central & inscribed angles · tangent ⊥ radius · radians & degrees · unit circle**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Read the center and radius from $(x - h)^2 + (y - k)^2 = r^2$, and write the equation from a center and radius, a diameter, or a graph.
- S2.** Complete the square to convert $x^2 + y^2 + ax + by + c = 0$ to standard form; use $r^2 > 0$ to find a constant.
- S3.** Translate or enlarge a circle and track the center and r^2 .
- S4.** Decide whether a point is on, inside or outside a circle; use tangency to an axis ($|\text{center coordinate}| = r$) or to another circle (distance between centers = $r_1 + r_2$ or $|r_1 - r_2|$).
- S5.** Compute arc length and sector area from a central angle in degrees or radians, and work backward to the radius or angle.
- S6.** Use the central-angle and inscribed-angle theorems (inscribed angle = $\frac{1}{2}$ central angle on the same arc; angle in a semicircle is 90° ; opposite angles of an inscribed quadrilateral are supplementary).
- S7.** Use “a tangent is perpendicular to the radius at the point of tangency”, including equal tangent segments from an outside point.
- S8.** Convert between radians and degrees; evaluate sin, cos, tan at standard angles on the unit circle and use quadrant signs.

2 · REQUIRED FACTS

- The graph of $(x - h)^2 + (y - k)^2 = r^2$ is a circle with center (h, k) and radius r (the right side is r^2 , not r).
- A point is inside the circle if its distance to the center is less than r , on it if equal, outside if greater.
- A circle is tangent to the x -axis when $|k| = r$, and to the y -axis when $|h| = r$.
- A full circle is $360^\circ = 2\pi$ radians, so $\pi \text{ rad} = 180^\circ$. An arc is the fraction $\frac{\theta}{360^\circ}$ (or $\frac{\theta}{2\pi}$) of the whole circle.
- An inscribed angle is half the central angle that intercepts the same arc. An angle inscribed in a semicircle is a right angle.
- A tangent line is perpendicular to the radius drawn to the point of tangency; two tangent segments from one outside point are equal.
- On the unit circle the point at angle θ is $(\cos \theta, \sin \theta)$; the signs follow the quadrant (All, Sine, Tangent, Cosine going counterclockwise).
- Common values: $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{\sqrt{2}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$; cos is the same list in reverse order.

3 · REQUIRED FORMULAS

$$(x - h)^2 + (y - k)^2 = r^2, \quad x^2 + y^2 + ax + by + c = 0 \Rightarrow \left(x + \frac{a}{2}\right)^2 + \left(y + \frac{b}{2}\right)^2 = c + \frac{a^2 + b^2}{4}$$

$$s = r\theta \ (\theta \text{ in radians}) = \frac{\theta^\circ}{360} \cdot 2\pi r, \quad A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{\theta^\circ}{360} \cdot \pi r^2 = \frac{1}{2}rs, \quad C = 2\pi r, \quad A = \pi r^2$$

$$\text{rad} = \text{deg} \cdot \frac{\pi}{180}, \quad \text{deg} = \text{rad} \cdot \frac{180}{\pi}, \quad \text{inscribed} = \frac{1}{2}\text{central}, \quad d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

4 · MINI THEORY SHEET

Equation of a circle. Group x -terms and y -terms, add $\left(\frac{\text{coefficient}}{2}\right)^2$ to *both sides* for each variable, then read h , k and r^2 . Signs flip: $(x+3)^2$ means $h = -3$. If a problem gives the radius, set the right-hand side equal to r^2 and solve for the constant.

Which formula for arcs and sectors?

Angle given in	Use
degrees	fraction $\frac{\theta}{360}$ of $2\pi r$ (arc) or of πr^2 (sector)
radians	$s = r\theta$ and $A = \frac{1}{2}r^2\theta$
arc and sector area known	$A = \frac{1}{2}rs$ gives r without knowing θ

Angles in a circle. Central angle = its arc (in degrees). Inscribed angle = $\frac{1}{2}$ its arc. Same arc $\Rightarrow$ equal inscribed angles. Draw radii to make isosceles triangles ($OA = OB = r$).

Tangents. Radius $\perp$ tangent gives a right triangle with legs r and the tangent length, and hypotenuse from the center to the outside point. Tangent lengths from one point are equal.

Radians. Convert with $\pi \leftrightarrow 180^\circ$. Co-terminal angles differ by 2π . A point at angle θ on a circle of radius r is $(r \cos \theta, r \sin \theta)$; moving clockwise subtracts the angle.

Speed shortcuts.

Situation	Shortcut
radius from $x^2 + y^2 + ax + by + c = 0$	$r^2 = \frac{a^2+b^2}{4} - c$
center from expanded form	$\left(-\frac{a}{2}, -\frac{b}{2}\right)$
circle tangent to an axis	the radius equals that center coordinate
diameter endpoints	center = midpoint, $r = \frac{1}{2} \cdot \text{distance}$
two sectors with equal area	$r_1^2\theta_1 = r_2^2\theta_2$
arc and radius given	$\theta = \frac{s}{r}$ rad, no degrees needed
tangent from an outside point	$\text{length}^2 = d^2 - r^2$
radius doubled/tripled	r^2 is multiplied by 4 or 9

5 · GUIDED EXAMPLES

Example 1 — Complete the square

The graph of $x^2 + y^2 + 6x - 8y - 11 = 0$ is a circle. Find its center and radius.

Group: $(x^2+6x)+(y^2-8y) = 11$. Add $\left(\frac{6}{2}\right)^2 = 9$ and $\left(\frac{8}{2}\right)^2 = 16$ to both sides: $(x+3)^2 + (y-4)^2 = 11+9+16 = 36$.
Center $(-3, 4)$, radius $\sqrt{36} = 6$.

Example 2 — Arc length and sector area

A circle has radius 9. A central angle measures 80° . Find the arc length and the sector area.

In radians, $80^\circ = 80 \cdot \frac{\pi}{180} = \frac{4\pi}{9}$. Arc: $s = r\theta = 9 \cdot \frac{4\pi}{9} = 4\pi$.

Sector: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(81)\frac{4\pi}{9} = 18\pi$. (Check: $\frac{80}{360} \cdot \pi(81) = 18\pi$.)

Example 3 — Tangent, radius and inscribed angle

$\overline{PA}$ is tangent to a circle with center O and radius 7 at A , and $OP = 25$. Also, an angle inscribed in the circle measures 35° . Find PA and the central angle on the same arc.

$OA \perp PA$, so $PA = \sqrt{25^2 - 7^2} = \sqrt{576} = 24$ (the triple 7-24-25).

Central angle $= 2 \cdot 35^\circ = 70^\circ$.

6 · PRACTICE SET — 30 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

The equation of a circle in the xy -plane is $(x - 3)^2 + (y + 5)^2 = 49$. Which of the following gives the center and the radius of the circle?

(A) center $(-3, 5)$, radius 49

(B) center $(3, -5)$, radius 7 ✓

(C) center $(-3, 5)$, radius 7

(D) center $(3, -5)$, radius 49

Worked solution

1. Compare with $(x - h)^2 + (y - k)^2 = r^2$: $h = 3$ and $k = -5$ (the sign flips), so the center is $(3, -5)$.

2. $r^2 = 49 \Rightarrow r = 7$.

Answer: (B)

2. EASY · NO CALCULATOR NEEDED

Which of the following is equal to $\frac{5\pi}{6}$ radians?

(A) 135°

(B) 300°

(C) 150° ✓

(D) 120°

Worked solution

1. Multiply by $\frac{180^\circ}{\pi}$: $\frac{5\pi}{6} \cdot \frac{180}{\pi} = 5 \cdot 30 = 150$.

2. So $\frac{5\pi}{6}$ radians $= 150^\circ$.

Answer: (C)

3. EASY · NO CALCULATOR NEEDED

A circle has a circumference of 20π . The area of the circle is $k\pi$. What is the value of k ?

Worked solution

1. $C = 2\pi r = 20\pi \Rightarrow r = 10$.

2. Area $= \pi r^2 = 100\pi$, so $k = 100$.

Answer: 100

4. EASY · NO CALCULATOR NEEDED

What is the value of $\sin \frac{5\pi}{6}$?

(A) $\frac{1}{2}$ ✓

(B) $-\frac{1}{2}$

(C) $\frac{\sqrt{3}}{2}$

(D) $-\frac{\sqrt{3}}{2}$

Worked solution

1. $\frac{5\pi}{6} = 150^\circ$, in Quadrant II where sine is positive.

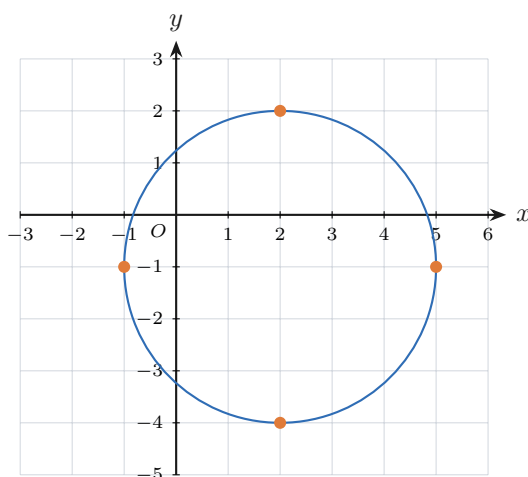
2. The reference angle is $180^\circ - 150^\circ = 30^\circ$, and $\sin 30^\circ = \frac{1}{2}$.

Answer: (A)

5.

EASY · DESMOS

The graph of a circle in the xy -plane is shown. Which of the following is an equation of the circle?



(A) $(x + 2)^2 + (y - 1)^2 = 9$

(B) $(x - 2)^2 + (y + 1)^2 = 3$

(C) $(x - 2)^2 + (y - 1)^2 = 9$

(D) $(x - 2)^2 + (y + 1)^2 = 9$ ✓

Worked solution

1. Read the horizontal diameter: the leftmost and rightmost grid points are $(-1, -1)$ and $(5, -1)$, so the center is $(2, -1)$ and the radius is 3.

2. $(x - 2)^2 + (y + 1)^2 = 3^2 = 9$.

Answer: (D)

6.

MEDIUM · NO CALCULATOR NEEDED

In the xy -plane, the graph of $x^2 + y^2 - 10x + 4y + 13 = 0$ is a circle. What is the radius of the circle?

Worked solution

1. Complete the square: $(x^2 - 10x + 25) + (y^2 + 4y + 4) = -13 + 25 + 4$.

2. $(x - 5)^2 + (y + 2)^2 = 16$.

3. $r = \sqrt{16} = 4$.

Answer: 4

7.

MEDIUM · NO CALCULATOR NEEDED

A circle has a radius of 12 centimeters. An arc of the circle has a central angle of 75° . What is the length, in centimeters, of the arc?

(A) 30π

(B) 75π

(C) 5π ✓

(D) 10π

Worked solution

1. Arc = $\frac{75}{360} \cdot 2\pi(12)$.

2. $\frac{75}{360} = \frac{5}{24}$, so the arc is $\frac{5}{24} \cdot 24\pi = 5\pi$.

Answer: (C)

8.

MEDIUM · NO CALCULATOR NEEDED

Which of the following points lies on the circle with equation $(x + 1)^2 + (y - 4)^2 = 25$?

(A) $(1, -4)$ (B) $(2, 8)$ ✓(C) $(-2, 7)$ (D) $(3, 0)$ **Worked solution**

1. Substitute each point; the sum of the squares must be 25.

2. $(2, 8)$: $(2 + 1)^2 + (8 - 4)^2 = 9 + 16 = 25$. ✓

3. $(-2, 7)$ gives $1 + 9 = 10$; $(3, 0)$ gives $16 + 16 = 32$; $(1, -4)$ gives $4 + 64 = 68$.

Answer: (B)

9.

MEDIUM · NO CALCULATOR NEEDED

A circular sector has a radius of 6 and a central angle of $\frac{2\pi}{3}$ radians. The area of the sector is $k\pi$. What is the value of k ?

Worked solution

$$1. A = \frac{1}{2}r^2\theta = \frac{1}{2}(36)\left(\frac{2\pi}{3}\right).$$

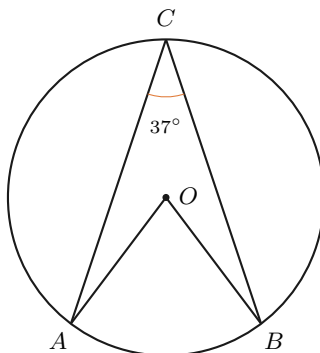
$$2. A = 18 \cdot \frac{2\pi}{3} = 12\pi, \text{ so } k = 12.$$

Answer: 12

10.

MEDIUM · NO CALCULATOR NEEDED

In the figure, A , B and C lie on the circle with center O . What is the measure, in degrees, of $\angle AOB$?



(A) 37

(B) 143

(C) 106

(D) 74 ✓

Worked solution

1. $\angle ACB$ is an inscribed angle and $\angle AOB$ is the central angle on the same arc AB .

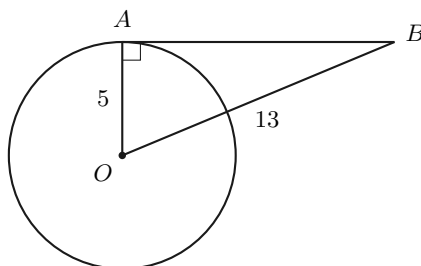
2. A central angle is twice the inscribed angle: $\angle AOB = 2(37^\circ) = 74^\circ$.

Answer: (D)

11.

MEDIUM · NO CALCULATOR NEEDED

In the figure, $\overline{AB}$ is tangent to the circle with center O at point A . What is the length of $\overline{AB}$?



(A) 12 ✓

(B) 8

(C) $\sqrt{194}$

(D) 18

Worked solution

1. A tangent is perpendicular to the radius at the point of tangency, so $\angle OAB = 90^\circ$.

2. $AB = \sqrt{OB^2 - OA^2} = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$.

Answer: (A)

12.

MEDIUM · NO CALCULATOR NEEDED

A circle has a radius of 24. An arc of the circle has length 10π . What is the measure, in degrees, of the central angle that intercepts this arc?

Worked solution

1. Radian measure: $\theta = \frac{s}{r} = \frac{10\pi}{24} = \frac{5\pi}{12}$.

2. Degrees: $\frac{5\pi}{12} \cdot \frac{180}{\pi} = 75$.

Answer: 75

13.

MEDIUM · NO CALCULATOR NEEDED

The endpoints of a diameter of a circle in the xy -plane are $(-2, 3)$ and $(6, -5)$. Which of the following is an equation of the circle?

(A) $(x - 2)^2 + (y + 1)^2 = 16$ (B) $(x - 2)^2 + (y + 1)^2 = 32$ ✓(C) $(x - 2)^2 + (y + 1)^2 = 128$ (D) $(x + 2)^2 + (y - 1)^2 = 64$ **Worked solution**

1. Center = midpoint = $\left(\frac{-2 + 6}{2}, \frac{3 + (-5)}{2}\right) = (2, -1)$.

2. Diameter length squared: $(6 - (-2))^2 + (-5 - 3)^2 = 64 + 64 = 128$, so $r^2 = \frac{128}{4} = 32$.

3. $(x - 2)^2 + (y + 1)^2 = 32$.

Answer: (B)

14.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the graph of $x^2 + y^2 - 8x - 6y + k = 0$ is a circle that is tangent to the x -axis, where k is a constant. What is the value of k ?

Worked solution

1. Complete the square: $(x - 4)^2 + (y - 3)^2 = 25 - k$. The center is $(4, 3)$.
 2. Tangent to the x -axis means the radius equals the center's distance to the axis: $r = 3$, so $r^2 = 9$.
 3. $25 - k = 9 \Rightarrow k = 16$.
- Answer:** 16

15.

HARD · NO CALCULATOR NEEDED

In the xy -plane, two different circles are each tangent to both coordinate axes, lie in Quadrant IV, and pass through the point $(9, -2)$. What is the sum of the radii of the two circles?

Worked solution

1. A circle in Quadrant IV tangent to both axes has center $(r, -r)$ and radius r : $(x - r)^2 + (y + r)^2 = r^2$.
 2. Substitute $(9, -2)$: $(9 - r)^2 + (-2 + r)^2 = r^2 \Rightarrow r^2 - 22r + 85 = 0$.
 3. The two roots are the two radii, $r = 5$ and $r = 17$ (check: $(9 - 5)^2 + (-2 + 5)^2 = 25$ and $(9 - 17)^2 + (-2 + 17)^2 = 289$). Their sum is $5 + 17 = 22$ (the sum of the roots of $r^2 - 22r + 85 = 0$).
- Answer:** 22

16.

HARD · NO CALCULATOR NEEDED

The circle $(x + 2)^2 + (y - 3)^2 = 20$ is translated 4 units to the left and 5 units down. Then the radius of the translated circle is tripled, with the center unchanged. Which of the following is an equation of the resulting circle?

- (A) $(x - 2)^2 + (y - 8)^2 = 180$ (B) $(x + 6)^2 + (y + 2)^2 = 540$
(C) $(x + 6)^2 + (y + 2)^2 = 180$ ✓ (D) $(x + 6)^2 + (y + 2)^2 = 60$

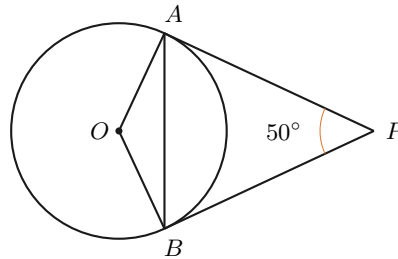
Worked solution

1. Translating left 4 and down 5 moves the center $(-2, 3)$ to $(-6, -2)$.
 2. Tripling the radius multiplies r^2 by $3^2 = 9$: $r^2 = 20 \cdot 9 = 180$.
 3. $(x + 6)^2 + (y + 2)^2 = 180$.
- Answer:** (C)

17.

HARD · NO CALCULATOR NEEDED

In the figure, $\overline{PA}$ and $\overline{PB}$ are tangent to the circle with center O at A and B , and $\angle APB = 50^\circ$. What is the measure, in degrees, of $\angle OAB$?



(A) 25 ✓

(B) 40

(C) 50

(D) 65

Worked solution

1. $\angle OAP = \angle OBP = 90^\circ$ (tangent $\perp$ radius). In quadrilateral $OAPB$: $\angle AOB = 360 - 90 - 90 - 50 = 130^\circ$.

2. $OA = OB$ (radii), so $\triangle OAB$ is isosceles with base angles $\angle OAB = \angle OBA$.

3. $\angle OAB = \frac{180 - 130}{2} = 25$.

Answer: (A)

18.

HARD · NO CALCULATOR NEEDED

A circular sector has an area of 40 square centimeters and an arc length of 10 centimeters. What is the radius, in centimeters, of the circle?

Worked solution

1. For a sector, $A = \frac{1}{2}rs$ (a triangle-like formula using the arc length s).

2. $40 = \frac{1}{2}r(10) = 5r \Rightarrow r = 8$.

3. Check: $\theta = \frac{s}{r} = \frac{10}{8} = 1.25$ rad, and $\frac{1}{2}r^2\theta = \frac{1}{2}(64)(1.25) = 40$.

Answer: 8

19.

HARD · NO CALCULATOR NEEDED

For an angle θ with $\pi < \theta < \frac{3\pi}{2}$, $\sin \theta = -\frac{5}{13}$. What is the value of $\tan \theta$?

(A) $-\frac{5}{12}$ (B) $\frac{12}{5}$ (C) $-\frac{12}{5}$ (D) $\frac{5}{12}$ ✓**Worked solution**

1. $\pi < \theta < \frac{3\pi}{2}$ is Quadrant III: both sine and cosine are negative.

2. $\cos \theta = -\sqrt{1 - \left(\frac{5}{13}\right)^2} = -\frac{12}{13}$.

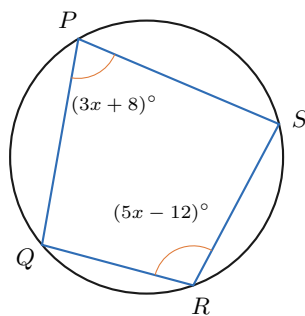
3. $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{-5/13}{-12/13} = \frac{5}{12}$ (positive in Quadrant III).

Answer: (D)

20.

HARD · NO CALCULATOR NEEDED

Quadrilateral $PQRS$ is inscribed in a circle. The measures of angles P and R are $(3x + 8)^\circ$ and $(5x - 12)^\circ$, respectively. What is the measure, in degrees, of angle R ?



(A) 115

(B) 77

(C) 103 ✓

(D) 23

Worked solution

- Angles P and R are opposite angles of an inscribed quadrilateral. Each intercepts one of the two arcs that together make 360° , so $\angle P + \angle R = \frac{360^\circ}{2} = 180^\circ$.
- $(3x + 8) + (5x - 12) = 180 \Rightarrow 8x - 4 = 180 \Rightarrow x = 23$.
- $\angle R = 5(23) - 12 = 103$. (Check: $\angle P = 3(23) + 8 = 77$ and $77 + 103 = 180$.)

Answer: (C)

21.

HARD · CALCULATOR

A circular running track has a radius of 0.12 kilometers. A runner starts at one point on the track and runs along the track through a central angle of 200° . To the nearest meter, how far does the runner run?

Worked solution

- Convert the radius: $0.12 \text{ km} = 120 \text{ m}$.
- Arc length $= \frac{200}{360} \cdot 2\pi(120) = \frac{5}{9} \cdot 240\pi = \frac{400\pi}{3} \approx 418.9 \text{ m}$.
- To the nearest meter: 419.

Answer: 419

22.

HARD · NO CALCULATOR NEEDED

A circular pizza with a radius of 9 inches is cut into slices, and one slice has a central angle of 40° . A second circular pizza has a radius of 12 inches. A slice of the second pizza has the same area as the 40° slice of the first pizza. What is the central angle, in degrees, of the slice of the second pizza?

(A) 71.1

(B) 22.5 ✓

(C) 30

(D) 53.3

Worked solution

- Sector area is $\frac{\theta}{360}\pi r^2$, so equal areas give $\theta_1 r_1^2 = \theta_2 r_2^2$.
- $40 \cdot 9^2 = \theta_2 \cdot 12^2 \Rightarrow \theta_2 = \frac{40 \cdot 81}{144}$.
- $\theta_2 = 22.5$.

Answer: (B)

23.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the graph of $x^2 + y^2 - 6x + 2ky - 39 = 0$ is a circle whose center lies on the line $y = x + 1$, where k is a constant. What is the radius of the circle?

Worked solution

1. Complete the square: $(x - 3)^2 + (y + k)^2 = 39 + 9 + k^2$. The center is $(3, -k)$.

2. The center is on $y = x + 1$: $-k = 3 + 1 \Rightarrow k = -4$.

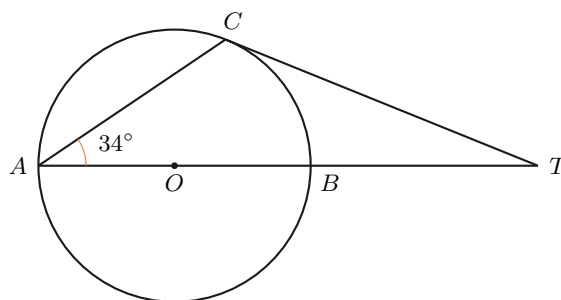
3. $r^2 = 39 + 9 + (-4)^2 = 64$, so $r = 8$.

Answer: 8

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, $\overline{AB}$ is a diameter of the circle with center O , point C lies on the circle, and line CT is tangent to the circle at C with T on line AB . What is the measure, in degrees, of $\angle CTA$?



(A) 34

(B) 56

(C) 68

(D) 22 ✓

Worked solution

1. $\angle CAB = 34^\circ$ is an inscribed angle on arc CB , so the central angle is $\angle COB = 2(34^\circ) = 68^\circ$.

2. The tangent is perpendicular to the radius: $\angle OCT = 90^\circ$.

3. In right triangle OCT : $\angle CTO = 90^\circ - 68^\circ = 22^\circ$.

4. $\angle CTA$ is the same angle as $\angle CTO$ (points A, O, T are collinear), so it measures 22° .

Answer: (D)

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

A circular sector has a perimeter of 20 centimeters (two radii plus the arc) and an area of 24 square centimeters. The central angle of the sector is θ radians. What is the sum of all possible values of θ ?

Worked solution

1. Let the radius be r and the arc be $r\theta$. Perimeter: $2r + r\theta = 20$, so the arc is $20 - 2r$.

2. Area = $\frac{1}{2}r(\text{arc}) = \frac{1}{2}r(20 - 2r) = 10r - r^2 = 24 \Rightarrow r^2 - 10r + 24 = 0 \Rightarrow r = 4$ or $r = 6$.

3. $\theta = \frac{20 - 2r}{r}$: for $r = 4$, $\theta = \frac{12}{4} = 3$; for $r = 6$, $\theta = \frac{8}{6} = \frac{4}{3}$.

4. Both are valid (each is less than 2π). Sum = $3 + \frac{4}{3} = \frac{13}{3}$.

Answer: $\frac{13}{3}$

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

The line $3x + 4y = k$, where k is a positive constant, is tangent to the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ in the xy -plane. What is the value of k ?

- (A) 19 ✓ (B) 25
(C) 31 (D) -31

Worked solution

1. Complete the square: $(x - 2)^2 + (y + 3)^2 = 12 + 4 + 9 = 25$. Center $(2, -3)$, radius 5.
2. The radius to a point of tangency is perpendicular to the tangent line. The line $3x + 4y = k$ has direction normal $(3, 4)$, so the radius points along $\pm(3, 4)/5$ and the tangent points are $(2, -3) \pm 5 \cdot \frac{(3, 4)}{5} = (5, 1)$ and $(-1, -7)$.
3. $k = 3(5) + 4(1) = 19$ or $k = 3(-1) + 4(-7) = -31$.
4. k is positive, so $k = 19$.

Answer: (A)

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

Circle A is given by $x^2 + y^2 - 2x - 4y - 11 = 0$ and circle B is given by $x^2 + y^2 - 14x + 12y + c = 0$ in the xy -plane, where c is a constant. The two circles are externally tangent (they touch at exactly one point and neither lies inside the other). What is the value of c ?

Worked solution

1. Circle A : $(x - 1)^2 + (y - 2)^2 = 16$, center $(1, 2)$, radius 4.
2. Circle B : $(x - 7)^2 + (y + 6)^2 = 85 - c$, center $(7, -6)$, radius $r_B = \sqrt{85 - c}$.
3. Distance between centers: $\sqrt{6^2 + 8^2} = 10$. External tangency: $10 = 4 + r_B \Rightarrow r_B = 6$.
4. $85 - c = 36 \Rightarrow c = 49$.

Answer: 49

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

A point (a, b) lies on the unit circle at angle θ radians in standard position, where $0 < \theta < 2\pi$ and $ab > 0$. Which of the following must be true?

I. $\tan \theta > 0$ II. $\sin(\theta + \pi) \cos(\theta + \pi) > 0$ III. $\sin \theta > 0$

- (A) I only (B) I and II only ✓
(C) II and III only (D) I, II, and III

Worked solution

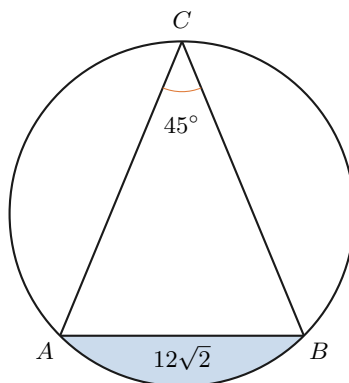
1. $a = \cos \theta$ and $b = \sin \theta$, and $ab > 0$ means sine and cosine have the same sign: Quadrant I or Quadrant III.
2. I: $\tan \theta = \frac{b}{a} > 0$ in both cases. True.
3. II: adding π negates both: $\sin(\theta + \pi) \cos(\theta + \pi) = (-b)(-a) = ab > 0$. True.
4. III: in Quadrant III, $\sin \theta < 0$, so III need not be true.
5. I and II only.

Answer: (B)

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, A , B and C lie on a circle, $\angle ACB = 45^\circ$, and $AB = 12\sqrt{2}$. The area of the shaded region, bounded by chord $\overline{AB}$ and the arc AB that does not contain C , is $a\pi - b$, where a and b are positive integers. What is the value of $a + b$?

**Worked solution**

1. The inscribed angle 45° gives the central angle $\angle AOB = 90^\circ$ (with center O).
2. $\triangle AOB$ is an isosceles right triangle with hypotenuse $AB = 12\sqrt{2}$, so $r\sqrt{2} = 12\sqrt{2}$ and $r = 12$.
3. Sector $AOB = \frac{90}{360}\pi(12)^2 = 36\pi$; triangle $AOB = \frac{1}{2}(12)(12) = 72$.
4. Shaded region $= 36\pi - 72$, so $a = 36$, $b = 72$ and $a + b = 108$.

Answer: 108

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

A point starts at $(6, 0)$ and moves clockwise along the circle $x^2 + y^2 = 36$ a distance of 11π . What are the coordinates of the point where it stops?

- (A) $(3\sqrt{3}, -3)$ (B) $(-3\sqrt{3}, 3)$
 (C) $(3\sqrt{3}, 3)$ ✓ (D) $(3, 3\sqrt{3})$

Worked solution

1. The central angle is $\theta = \frac{s}{r} = \frac{11\pi}{6}$ radians, measured clockwise, i.e. $-\frac{11\pi}{6}$ in standard position.
2. $-\frac{11\pi}{6} + 2\pi = \frac{\pi}{6}$, so the final position is at angle 30° .
3. $(x, y) = \left(6 \cos \frac{\pi}{6}, 6 \sin \frac{\pi}{6}\right) = \left(6 \cdot \frac{\sqrt{3}}{2}, 6 \cdot \frac{1}{2}\right) = (3\sqrt{3}, 3)$.

Answer: (C)